

COVID-19 Patient Data Analysis Using Python

Minor Project Report

Submitted By

Harshita Singh

B.Tech (AI)

Submitted To

Ediglobe

INTRODUCTION

The COVID-19 pandemic affected millions of people worldwide and generated a large amount of healthcare data. Analyzing this data helps in understanding infection patterns, patient demographics, and regional trends. Exploratory Data Analysis (EDA) is an important step in data analysis that helps identify patterns, detect anomalies, and gain meaningful insights before applying advanced analytical techniques.

This project focuses on the analysis of a COVID-19 patient dataset using Python. Various Python libraries such as Pandas, NumPy, and Matplotlib were used for data cleaning, pre-processing, visualization, and analysis. Different charts and graphs were created to understand patient distribution, infection reasons, regional trends, age distribution, and correlations among different variables. The findings from this analysis provide a better understanding of the dataset and demonstrate the practical application of Exploratory Data Analysis techniques.

OBJECTIVES

- To analyze the COVID-19 patient dataset using Python.
- To perform data cleaning and preprocessing for accurate analysis.
- To conduct Exploratory Data Analysis (EDA) for identifying meaningful patterns and trends.
- To visualize the data using charts and graphs for better understanding.
- To study the relationship between different variables using correlation analysis.
- To gain practical experience in data analysis using Python libraries such as Pandas, NumPy, and Matplotlib .

TOOLS & TECHNOLOGIES USED

Tool/Technology	Purpose
Python	Programming Language
Pandas	Data Cleaning & Analysis
NumPy	Numerical Computation
Matplotlib	Data Visualization
Jupyter Notebook	Development Environment

DATASET DESCRIPTION

The dataset used in this project contains information about COVID-19 patients, including demographic details, infection sources, regional information, contact history, and patient status. The dataset was collected for educational and analytical purposes to perform Exploratory Data Analysis (EDA). Before analysis, the data was cleaned and preprocessed by handling missing values, removing unnecessary columns, and converting data into an appropriate format. This prepared dataset was then used to generate meaningful visualizations and insights.

Dataset Features

- Patient ID
- Age
- Birth Year
- Gender
- Province/Region
- Infection Reason
- Infection Order
- Contact Number
- Confirmed Date
- Patient Status

METHODOLOGY

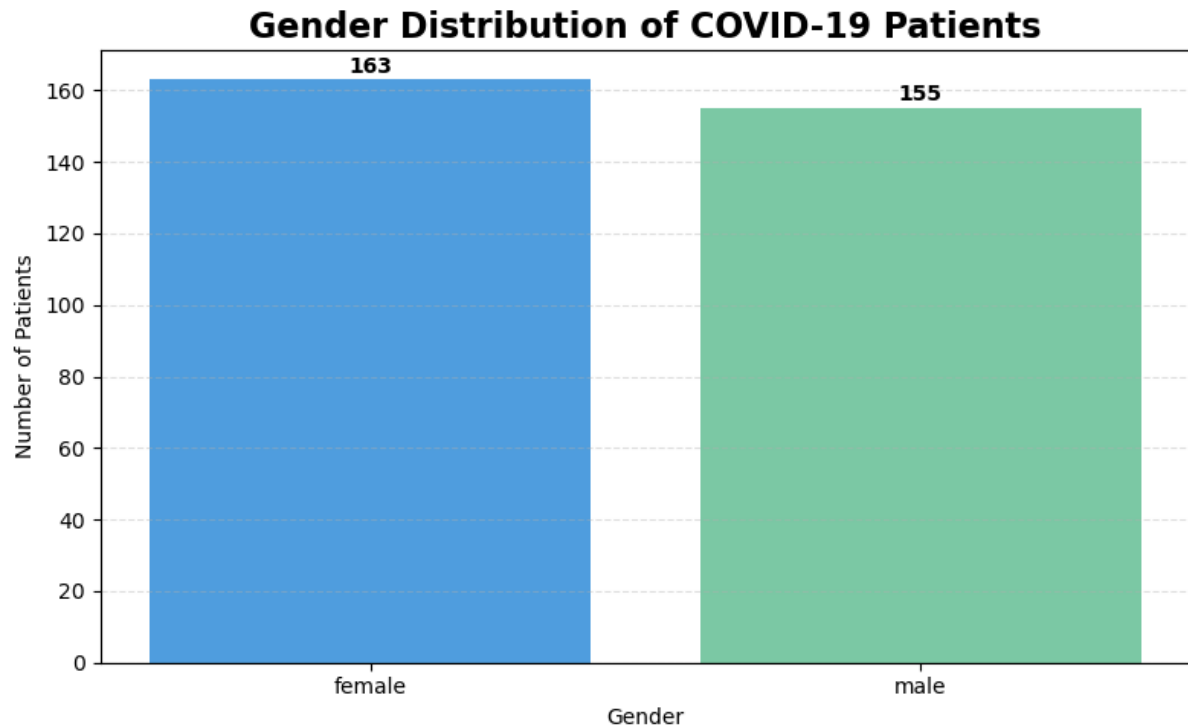
The methodology followed in this project began with loading the COVID-19 patient dataset using the Pandas library. The data was preprocessed by checking missing values, removing duplicates where needed, and formatting columns appropriately. Exploratory Data Analysis was performed using Pandas, NumPy, and Matplotlib to generate visualizations.

- Data collection
- Data Cleaning
- Data Pre-processing
- Exploratory Data Analysis
- Data Visualization
- Interpretation of Result

EXPLORATORY DATA ANALYSIS (EDA)

Exploratory Data Analysis (EDA) was performed to understand the characteristics of the COVID-19 patient dataset. Different visualization techniques were used to identify trends, patterns, and relationships among the variables. Charts and graphs helped analyze patient distribution, infection reasons, regional trends, and correlations between numerical features. The insights obtained from EDA improved the understanding of the dataset and supported meaningful interpretation of the data.

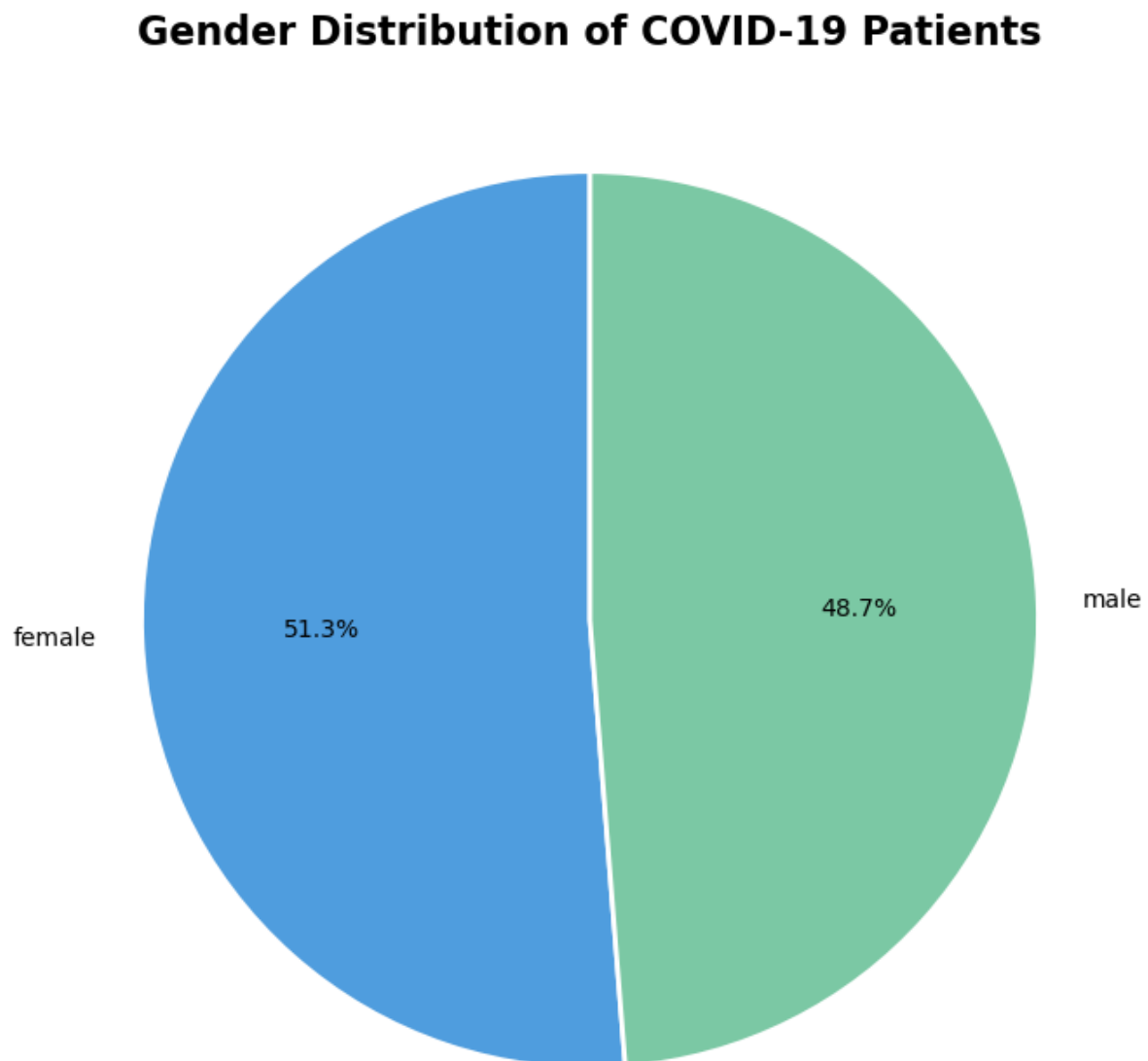
Figure 1: Gender Distribution of COVID-19 Patients (Bar Graph)



Observation:

The pie chart represents the percentage distribution of COVID-19 patients by gender. Female patients account for **51.3%** of the total cases, while male patients account for **48.7%**. This indicates that the gender distribution in the dataset is nearly balanced.

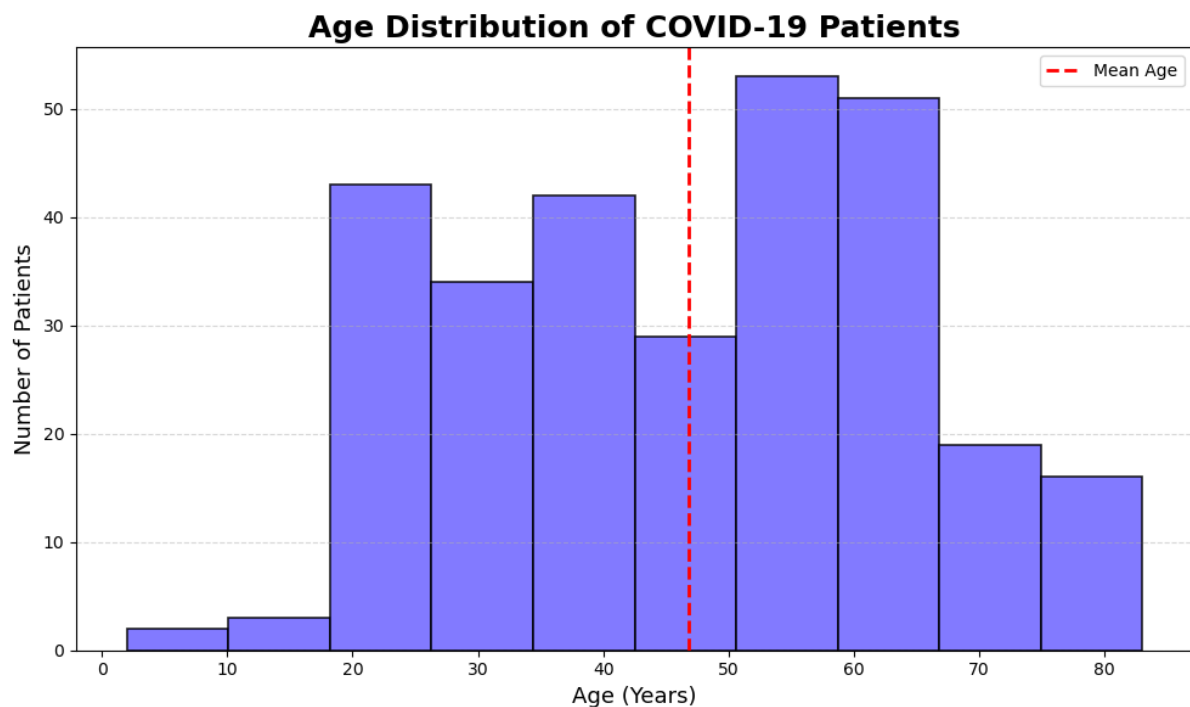
Figure 2: Gender Distribution of COVID-19 Patients (Pie Chart)



Observation:

The pie chart represents the percentage distribution of COVID-19 patients by gender. Female patients account for **51.3%** of the total cases, while male patients account for **48.7%**. This indicates that the gender distribution in the dataset is nearly balanced.

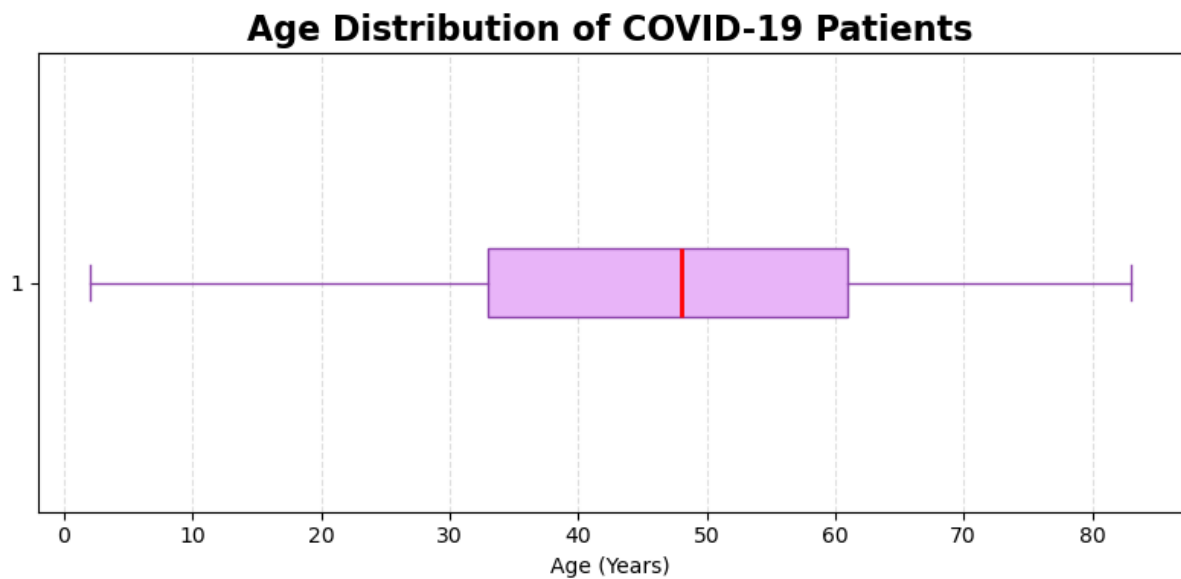
Figure 3: Age Distribution of COVID-19 Patients (Histogram)



Observation:

The histogram shows the distribution of COVID-19 patients across different age groups. Most patients are concentrated in the **middle-aged and older age groups (approximately 50–65 years)**, while comparatively fewer patients belong to the very young or very old age groups. The red dashed line indicates the **mean age**, which is around **47 years**.

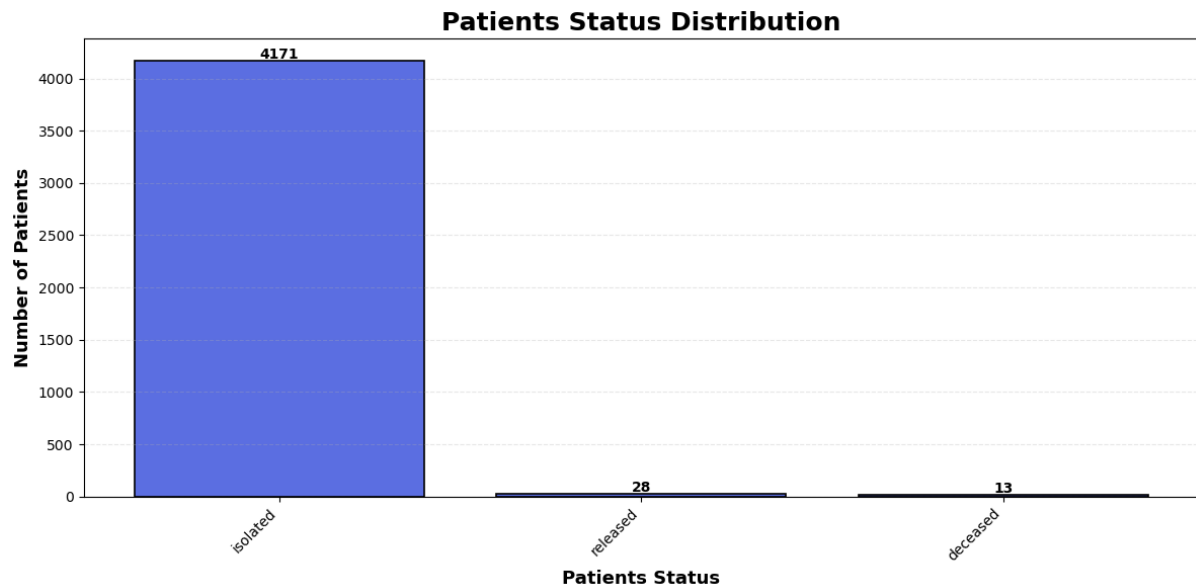
Figure 4: Age Distribution of COVID-19 Patient (Box Plot)



Observation:

The box plot summarizes the distribution of patients' ages. The **median age is around 48 years**, and most patients fall within the **interquartile range of approximately 33 to 61 years**. The plot does not indicate any significant outliers, suggesting that the age values are fairly well distributed.

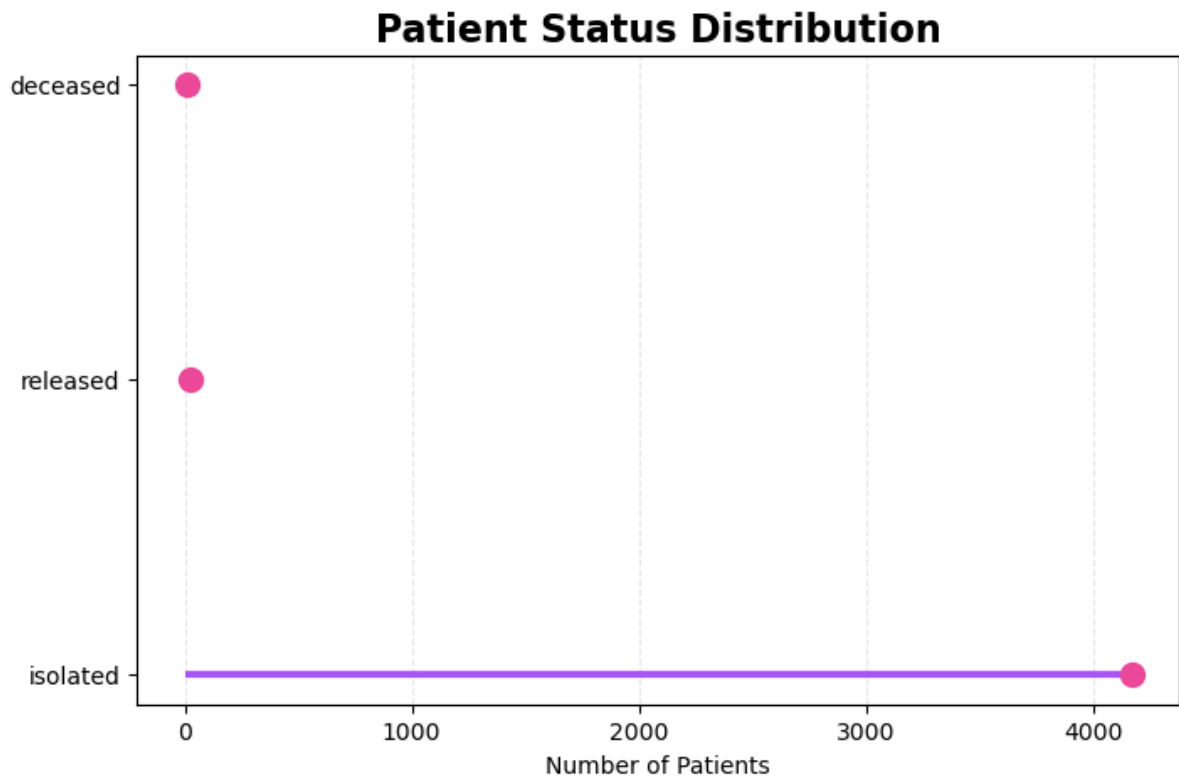
Figure 5: Patient Status Distribution



Observation:

The bar graph shows the distribution of patients based on their COVID-19 status. Most patients in the dataset are **isolated (4,171)**, while only **28 patients were released** and **13 patients were deceased**. This indicates that the majority of records belong to patients who were in isolation at the time the dataset was collected.

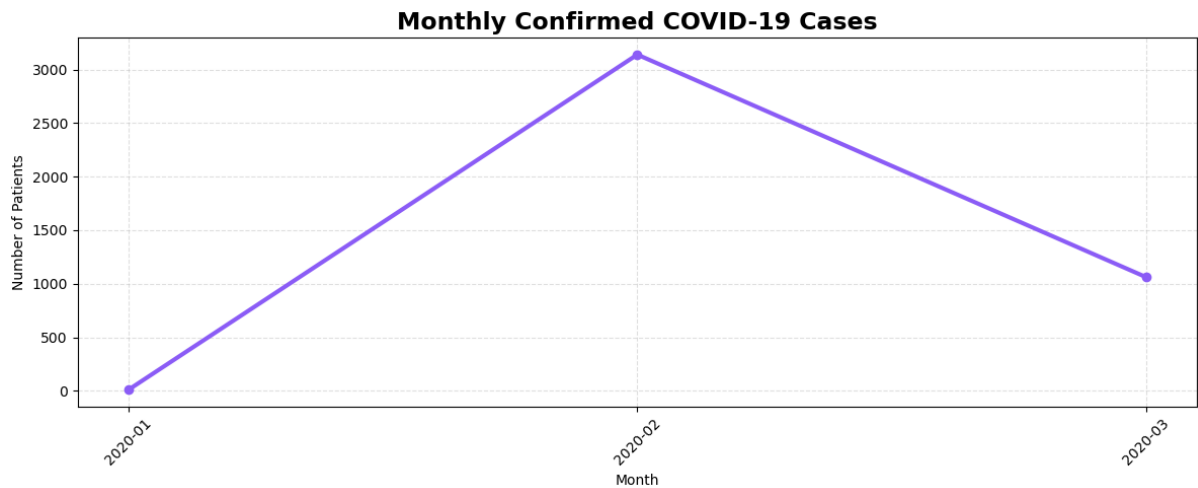
Figure 6: Patient Status Distribution (Dot Plot)



Observation:

The dot plot presents the distribution of patients according to their COVID-19 status. The **isolated** category has the highest number of patients (**4,171**), whereas the **released** (**28**) and **deceased** (**13**) categories contain significantly fewer patients. This clearly indicates that the majority of patient records belong to individuals who were under isolation.

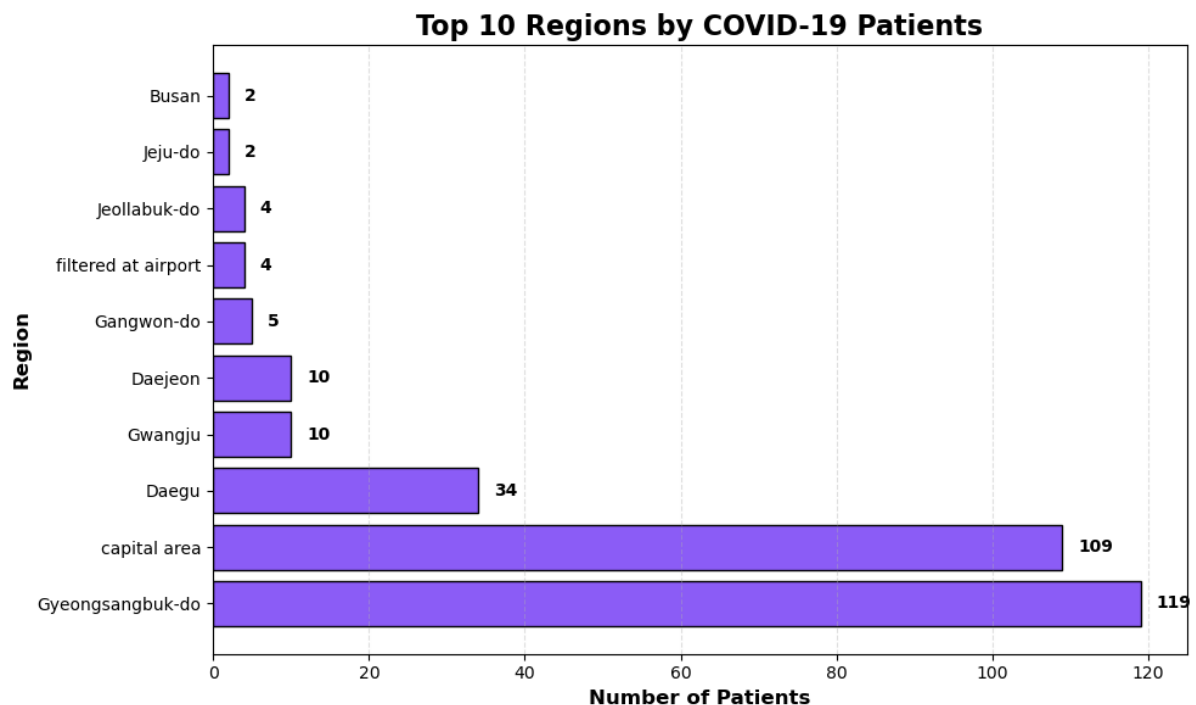
Figure 7: Monthly Confirmed COVID-19 Cases



Observation:

The line graph shows the monthly trend of confirmed COVID-19 cases in the dataset. The number of confirmed cases increased sharply from **January 2020** and reached its peak in **February 2020**. After February, the number of confirmed cases declined in **March 2020**, indicating a decrease in the recorded cases during that period.

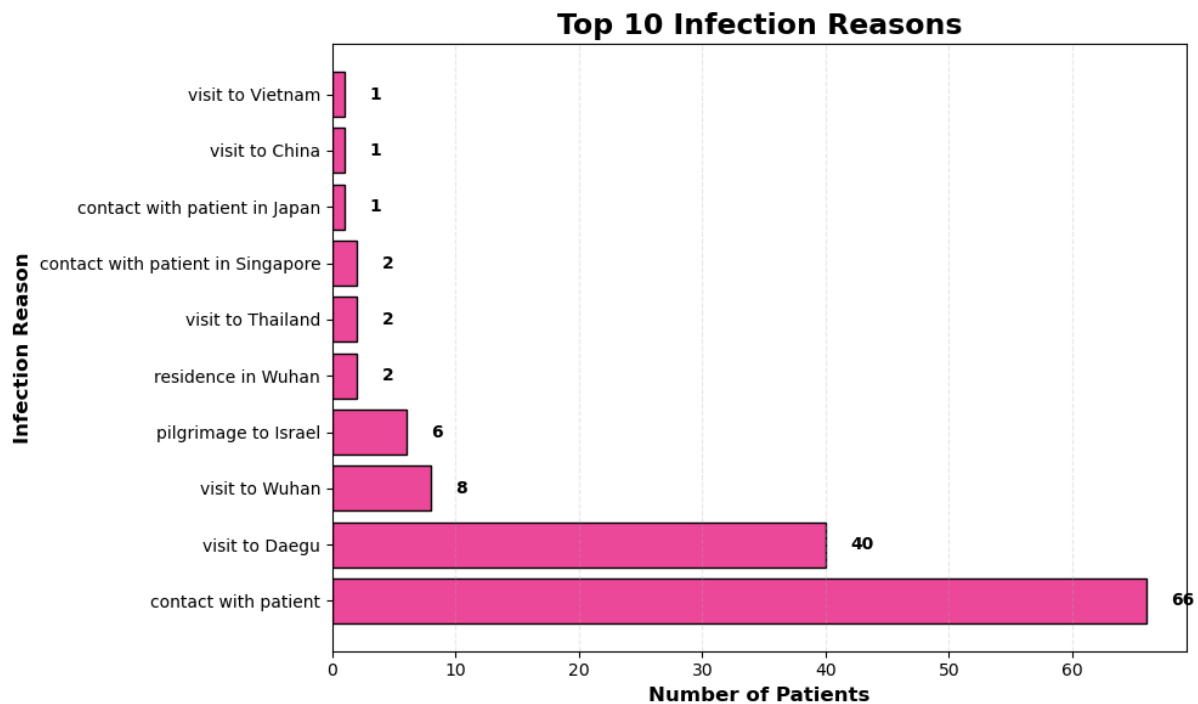
Figure 8: Top 10 Regions by COVID-19 Patients



Observation:

The horizontal bar chart shows the top 10 regions with the highest number of COVID-19 patients. **Gyeongsangbuk-do** recorded the highest number of patients (**119**), followed by the **Capital Area (109)** and **Daegu (34)**. Other regions such as Gwangju, Daejeon, Gangwon-do, Jeollabuk-do, Jeju-do, and Busan reported comparatively fewer cases. This indicates that COVID-19 cases were concentrated in a few major regions.

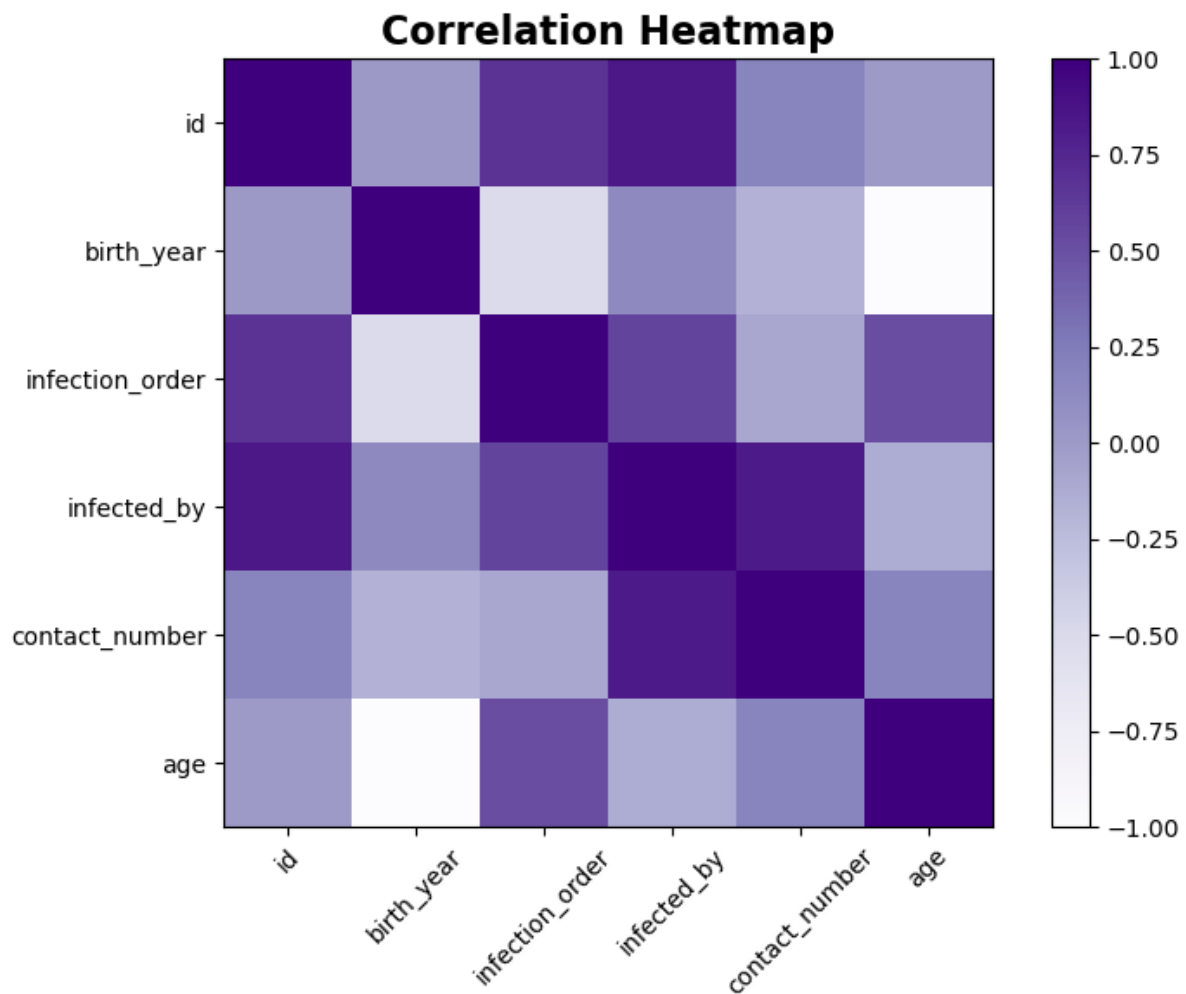
Figure 9: Top 10 Infection Reasons



Observation:

The graph shows the most common reasons for COVID-19 infection among patients. **Contact with an infected patient** was the leading cause (66 cases), followed by **visit to Daegu** (40 cases) and **visit to Wuhan** (8 cases). Other reasons, such as **pilgrimage to Israel**, **residence in Wuhan**, and visits to countries like **Thailand, China, and Vietnam**, contributed relatively fewer cases. This indicates that direct contact with infected individuals was the primary source of infection in the dataset.

Figure 10: Correlation Heatmap



Observation:

The correlation heatmap illustrates the relationships among the numerical variables in the dataset. Most variables show **weak to moderate correlations**, indicating that they are not strongly dependent on each other. A positive correlation is observed between some variables such as **infection order** and **infected by**, while **birth year** and **age** show a negative relationship, as expected. The heatmap provides a quick overview of how different numerical features are related.

Conclusion

This project focused on analyzing the COVID-19 patient dataset using Python for data cleaning, preprocessing, and exploratory data analysis (EDA). Various visualization techniques such as bar charts, pie charts, histograms, box plots, line graphs, horizontal bar charts, dot plots, and heatmaps were used to understand the characteristics of the dataset.

The analysis revealed important insights regarding gender distribution, age distribution, patient status, regional spread, infection reasons, monthly confirmed cases, and relationships among numerical variables. The findings show that most patients were isolated, middle-aged individuals formed the largest group, and direct contact with infected patients was the most common source of infection.

Overall, the project successfully demonstrated how Python-based data analysis techniques can transform raw healthcare data into meaningful insights, supporting better understanding and interpretation of COVID-19 patient information.

Future Scope

This project can be further enhanced by using larger and more recent COVID-19 datasets to obtain more comprehensive insights. Machine Learning algorithms can be applied to predict patient outcomes, infection risks, and recovery rates. Interactive dashboards can also be developed using tools such as Power BI, Tableau, or Streamlit for real-time visualization. In addition, integrating live healthcare data and advanced analytics techniques can improve decision-making and support future healthcare research.

References

- 1) Python Software Foundation. *Python Documentation*.
<https://docs.python.org/>
- 2) Pandas Development Team. *Pandas Documentation*.
<https://pandas.pydata.org/docs/>
- 3) NumPy Developers. *NumPy Documentation*. <https://numpy.org/doc/>
- 4) Matplotlib Development Team. *Matplotlib Documentation*.
<https://matplotlib.org/stable/>
- 5) Seaborn Documentation. <https://seaborn.pydata.org/>
- 6) COVID-19 Patient Dataset (Dataset used in this project).
- 7) GitHub Repository: **COVID-19-Patient-Data-Analysis-Using-Python**

GitHub Repository: GitHub Repository:

<https://github.com/ArpitaSingh2005/COVID-19-Patient-Data-Analysis-Using-Python>

